

YOLOv26 Model Evaluation and Comparative Analysis

1. Introduction

This report evaluates three YOLOv26 models trained with different optimizers and hyperparameters. Performance is assessed using confusion matrices and statistical metrics to determine deployment readiness.

2. Model Configurations

Model	Optimizer	Epochs	Batch	Learning Rate
Model 1	AdamW	25	4	0.01
Model 2	SGD	30	20	0.001
Model 3	Auto (SGD)	40	Auto	0.0001

3. Confusion Matrix Analysis (Model 3)

Model 3 demonstrates the most balanced confusion matrix. SUV true positives reach 408, while car detection improves significantly. Bicycle shows partial detection (TP=5), making Model 3 the only configuration to learn this rare class.

Although background misclassifications remain, predictions are more evenly distributed along the diagonal, reflecting improved decision boundaries due to extended training.

4. Aggregate Performance Metrics

Model	Precision	Recall	mAP50	mAP50-95	F1 Score
Model 1	0.7202	0.4010	0.4573	0.2611	0.5151
Model 2	0.5843	0.3525	0.3471	0.1971	0.4397
Model 3	0.6583	0.5072	0.5502	0.3316	0.5729

5. Per-Class Performance Summary

Bicycle consistently underperformed across models due to severe class imbalance. Model 3 achieved partial learning (mAP50=0.430), while Models 1 and 2 had zero recall.

SUV was the strongest class across all models. Motorcycle and Human showed moderate but unstable detection due to occlusion and scale variance.

6. Metric Interpretation

- Precision: Model 1 highest but overly selective
 - Recall: Model 3 highest, fewer missed detections
 - mAP50: Model 3 strongest overall localization accuracy
 - F1 Score: Confirms Model 3 as most balanced
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7. Deployment Recommendation

Model 3 is the most suitable for deployment due to:

- Highest mAP50, Recall, and F1 Score
- Partial detection of minority classes
- Balanced precision-recall tradeoff

Model 1 is unsuitable for safety-critical deployment due to high false negatives. Model 2 underperformed across all metrics.

8. Future Improvements

- Augment minority classes
 - Extend training to 60–80 epochs
 - Apply learning rate scheduling
 - Evaluate on a held-out test set
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9. Conclusion

This study demonstrates the importance of optimizer selection, learning rate control, and training duration in YOLOv26 performance. Model 3 achieved the most deployment-ready configuration.